

A Product Extension of the c_0 Topological CPCP/SR Separation

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2026-06-20

Status

This is a partial result for the broad question raised by Lopez-Perez and Medina in [1]: starting with a Banach space failing the classical CPCP, can one build a locally convex topology on its unit ball, containing the weak topology, such that the unit ball still fails topological CPCP but satisfies topological strong regularity?

The source paper answers this for B_{c_0} . The result below extends the same separation to every ℓ_∞ -sum

$$X = c_0 \oplus_\infty Y$$

with Y separable.

Source Question and Input Theorem

The source paper describes the broad strategy as follows.

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B_X , containing the weak topology so that B_X still fails τ -CPCP but satisfies τ -SR? Notice that a positive answer to this question would show that RNP and KMP are indeed equivalent. We don't know the answer to this question, but a first step is to answer the above question for the unit ball of c_0 , B_{c_0} , since it fails CPCP. In fact, every nonempty weakly open subset of B_{c_0} has diameter 2, that is B_{c_0} enjoys the D δ P for $\delta = 2$. The interesting fact is that B_{c_0} also fails to be SR even for open sets so that the weak topology does not make the trick.

In this section we give a positive answer to the latter question for B_{c_0} and for that purpose we state and prove the following lemma.

Lemma 3.1. Let $k \in \mathbb{N}$, $\mathcal{A} \subset \mathcal{P}(\mathbb{N})$ countable family and $A_n \in \mathcal{A}$. There is a
Its key c_0 theorem is [1, Theorem 3.2].

Now, we pass to show the existence of an appropriate topology on B_{c_0} . From now on we say that an interval $I \subset [-1, 1]$ is a proper open subinterval whenever it is open as a subset of $[-1, 1]$ and different from $[-1, 1]$.

Theorem 3.2. There exists a locally convex topology τ in B_{c_0} containing the weak topology such that B_{c_0} is τ -strongly regular for open subsets, but B_{c_0} has the τ -Diameter δ Property for $\delta = 2$.

We use the theorem in the following form. There is a countably based locally convex topology σ on B_{c_0} , containing the weak topology, such that:

1. every nonempty σ -open subset of B_{c_0} has norm diameter 2;
2. for every nonempty σ -open subset $U \subset B_{c_0}$ and every $\varepsilon > 0$, U contains a convex combination of nonempty relatively σ -open subsets whose diameter is less than ε .

The second item is slightly stronger than the formal “SR for open subsets” definition, and is exactly what is proved in the last paragraph of [1, Theorem 3.2].

Proof Intuition

The source construction makes the c_0 -coordinate simultaneously large and small in two different senses: every open set leaves infinitely many coordinates free, forcing diameter 2, while a later refinement of the subbase supplies small convex averages inside each open set.

In an ℓ_∞ -sum $c_0 \oplus_\infty Y$, the c_0 -coordinate still controls the large-diameter obstruction. To control convex combinations, put the norm topology on the Y -coordinate. Then inside any product neighborhood, choose the source paper’s small convex average in c_0 and a norm-small convex neighborhood in Y . The ℓ_∞ -norm takes the maximum of the two diameters, so both estimates survive.

Main Result

Theorem 1. *Let Y be a separable Banach space and let*

$$X = c_0 \oplus_\infty Y.$$

Then there is a countably based locally convex topology τ on B_X , containing the weak topology of B_X , such that:

1. *every nonempty τ -open subset of B_X has diameter 2;*
2. *B_X is τ -strongly regular for open subsets.*

In particular, B_X fails τ -CPCP while satisfying the topological strong-regularity conclusion for open subsets.

Proof. Since $X = c_0 \oplus_\infty Y$, its closed unit ball is the product

$$B_X = B_{c_0} \times B_Y$$

with norm

$$\|(x, y)\| = \max\{\|x\|_\infty, \|y\|\}.$$

Let σ be the topology on B_{c_0} supplied by [1, Theorem 3.2]. Define τ on B_X to be the product of σ on B_{c_0} and the norm topology on B_Y . Since Y is separable and σ has a countable base, τ has a countable base. It is locally convex. It contains the weak topology on B_X , because

$$X^* = (c_0 \oplus_\infty Y)^* = \ell_1 \oplus_1 Y^*,$$

so the weak topology on $B_X = B_{c_0} \times B_Y$ is the product of the weak topologies on the two coordinates; σ contains the weak topology on B_{c_0} , and the norm topology on B_Y contains the weak topology.

Let O be a nonempty τ -open subset of B_X . Choose a nonempty basic product $U \times V \subset O$, where $U \subset B_{c_0}$ is σ -open and $V \subset B_Y$ is norm-open. By the source theorem, $\text{diam}(U) = 2$. For every $\eta > 0$, choose $u_1, u_2 \in U$ with $\|u_1 - u_2\|_\infty > 2 - \eta$, and choose any $v \in V$. Then

$$\|(u_1, v) - (u_2, v)\| = \|u_1 - u_2\|_\infty > 2 - \eta.$$

Since the diameter of B_X is 2, $\text{diam}(O) = 2$. Thus B_X fails τ -CPCP already with the convex set $C = B_X$.

It remains to prove strong regularity for open subsets. Let O be a nonempty convex τ -open subset of B_X , and let $\varepsilon > 0$. Choose a nonempty product $U \times V \subset O$. Shrinking if necessary, assume V is convex and has norm diameter less than ε . By the source theorem, there are nonempty relatively σ -open subsets $U_1, \dots, U_m$ of U and positive coefficients $\lambda_1, \dots, \lambda_m$ summing to 1 such that

$$\text{diam} \left(\sum_{i=1}^m \lambda_i U_i \right) < \varepsilon.$$

For each i , set $O_i = U_i \times V$. These are nonempty relatively τ -open subsets of O . Since V is convex,

$$\sum_{i=1}^m \lambda_i O_i = \left(\sum_{i=1}^m \lambda_i U_i \right) \times V.$$

Therefore

$$\text{diam} \left(\sum_{i=1}^m \lambda_i O_i \right) = \max \left\{ \text{diam} \left(\sum_{i=1}^m \lambda_i U_i \right), \text{diam}(V) \right\} < \varepsilon.$$

This is precisely τ -strong regularity for open subsets. □

Corollary 1. *For every separable Banach space Y , the unit ball of $c_0 \oplus_\infty Y$ has a locally convex topology containing the weak topology for which the topological diameter two property holds while the corresponding strong diameter two property fails.*

Proof. The theorem gives diameter 2 for every nonempty τ -open subset. It also gives, inside every nonempty convex τ -open subset and for every $\varepsilon > 0$, a convex combination of relatively τ -open subsets with diameter less than ε . This is the announced separation. □

Limitations

This does not solve the full question from [1]: it covers spaces with a literal ℓ_∞ -summand c_0 , not arbitrary Banach spaces failing CPCP and not arbitrary spaces merely containing an isomorphic copy of c_0 . The ℓ_∞ -sum norm is used essentially to preserve the diameter estimate in the c_0 -coordinate and the small-diameter estimate in the Y -coordinate by taking a maximum.

Novelty Check

Local run indexes were searched for arXiv:2305.18976, CPCP, strong regularity, Krein–Milman property, and related RNP/KMP terms; no existing packet for this target was found. Web searches on 2026-06-20 for exact and close phrases including "2305.18976" "c_0" "locally convex topology",

"CPCP" "strong regularity" " c_0 " "locally convex topology", and " τ -CPCP" " τ -SR" " c_0 " did not locate a later statement of this $c_0 \oplus_\infty Y$ extension. The theorem is best viewed as a short new product permanence observation built from [1, Theorem 3.2].

Human Review Recommendation

Recommended as a likely valid partial result. The main point to verify is that the product topology $\sigma \times \|\cdot\|$ is an admissible locally convex topology on $B_{c_0 \oplus_\infty Y}$ containing the weak topology, and that the product-diameter calculation exactly matches the source definitions of τ -CPCP and τ -SR for open subsets.

References

- [1] G. Lopez-Perez and R. Medina, *The equivalence between CPCP and strong regularity under Krein-Milman property*, arXiv:2305.18976, 2023. <https://arxiv.org/abs/2305.18976>